



Documind AI

Enterprise AI Platform for Chatting, Editing, and Collaborating on PDF Documents

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Agenda

- 01 Problem Statement
- 02 Solution Overview
- 03 Technology Stack
- 04 System Architecture
- 05 Core Features
- 06 Standout Innovations
- 07 Security, Privacy & Billing
- 08 Challenges & Key Learnings
- 09 Live Demo

THE PROBLEM

Working with PDFs is Still Slow and Fragmented

Information overload

Long PDFs (contracts, papers, reports) take hours to read and cross-reference manually.

Scattered tooling

Summarizing, editing, e-signing, and quizzing on a document each require separate, disconnected apps.

No real collaboration

Teams email PDFs back and forth with no shared, live workspace to discuss them together.

Trust gap in AI answers

Generic AI chatbots give answers with no way to verify they're actually grounded in the source.

THE SOLUTION

Documind AI: One Platform, Every Document Workflow



One platform to chat, summarize, quiz, edit, sign, redact, and collaborate on any PDF.



Every AI answer is grounded with citations you can click to jump straight to the source page.



Built-in team workspaces with live collaboration, not just file sharing.



A public API so the same AI capabilities can be scripted and integrated elsewhere.

Technology Stack

FRONTEND

React 19, TypeScript, Vite, Tailwind CSS v4, Framer Motion

AI / ML

Google Gemini (gemini-3.5-flash), LangChain, FAISS vector search, gemini-embedding-2

BILLING

Stripe subscriptions

PDF ENGINE

pdf-lib, pdfjs-dist (client-side rendering & editing)

BACKEND

FastAPI (Python), SQLAlchemy, WebSockets

DATABASE

PostgreSQL

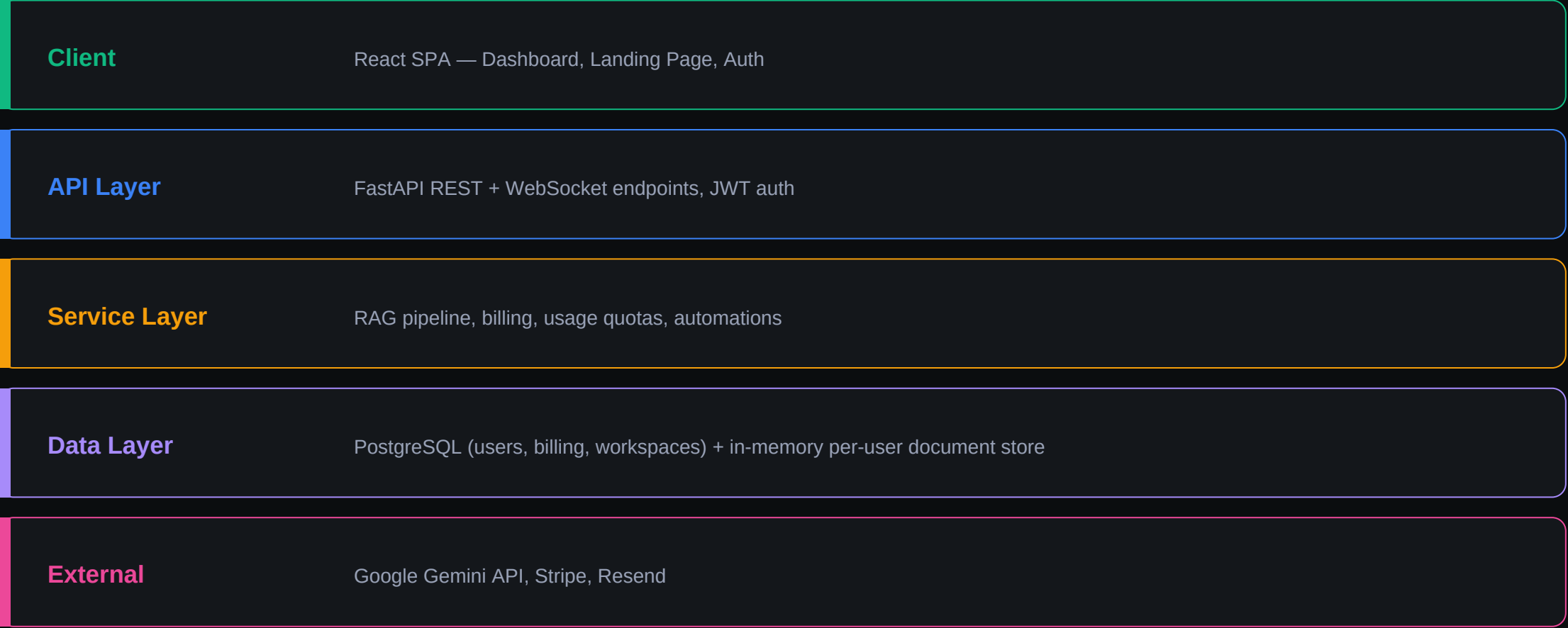
EMAIL

Resend (transactional email)

DEPLOYMENT

Render (backend, Docker) + Vercel (frontend)

System Architecture



Core Features

Chat with Citations

Ask questions across multiple PDFs with clickable, page-accurate citations.

Summaries & Study Notes

One-click structured summaries and exam-ready study notes.

Quiz & Flashcards

Auto-generated multiple-choice quizzes and flashcards from any document.

Document Compare

Side-by-side AI comparison of two documents' similarities and differences.

Research Assistant

Decomposes a topic into sub-questions and synthesizes a cited research report.

PDF Editor Suite

Click-to-edit text, e-signatures, form filling, OCR, AI rewriting, PII auto-redaction.

Batch Processing

Run one action across many documents at once.

AI Agent

Describe a goal in plain English; the agent plans and executes multi-step tasks.

Automations

Trigger-based rules that auto-run AI actions whenever a matching file is uploaded.

WHAT MAKES IT DIFFERENT

Standout Innovations

AI Audio Overview

Generates a natural two-host podcast-style conversation discussing the document, played back with distinct AI voices — comparable to NotebookLM's audio overviews.

Knowledge Graph

Extracts key entities and relationships from a document and renders an interactive, color-coded, force-directed graph — built with a custom physics simulation, no charting library.

Live Collaborative Cursors

Real-time WebSocket presence in Team Workspaces — see teammates' cursors move live on a shared PDF, Google-Docs style.

Click-to-Locate Citations

Every chat citation is clickable — jumps the PDF viewer straight to the cited page and flashes a highlight.

Public API

Pro users generate personal access keys and call chat, summarization, and generation endpoints from their own code.

Security, Privacy & Billing

Privacy-first storage	Uploaded documents live in memory for your session only — never written to disk unless you explicitly save a version or share to a workspace.
Secure authentication	Bcrypt password hashing, JWT access + refresh tokens, email verification, rate-limited password reset.
Encrypted in transit	HTTPS everywhere, browser-only Gemini API key storage — never sent to Documind's own servers unless required for generation.
Stripe-managed billing	Card details are handled entirely by Stripe; Documind never stores payment information.
Tiered usage quotas	Free tier: 15 actions/day, 2 documents. Pro tier (\$4.99/mo): unlimited actions, Team Workspaces, Public API.

Challenges & Key Learnings

Mid-project model deprecation

Google deprecated gemini-2.5-flash and gemini-embedding-001 for new API keys mid-development. Required auditing every AI call site and migrating to gemini-3.5-flash / gemini-embedding-2 without breaking existing sessions.

Real-time collaboration

Building live cursor presence required introducing WebSockets into a purely REST API, with in-memory per-workspace room management and JWT auth over a WebSocket handshake.

Privacy vs. collaboration

The core design promise was 'documents are never stored' — Team Workspaces and Version History required carefully scoping persistent storage as an explicit, disclosed opt-in exception rather than breaking that promise silently.

Full internationalization

Supporting English, French, Arabic, and Russian — including right-to-left layout for Arabic — across every panel in the app, not just the landing page.

LIVE DEMONSTRATION

Let's See It in Action

Chat -> Audio Overview -> Knowledge Graph -> Live Collaboration

<https://documindai.online>

Thank You

Questions & Discussion

LIVE APPLICATION

<https://documindai.online>

SOURCE CODE

<https://github.com/Abdullah001butt/rag-pdf-chatbot>